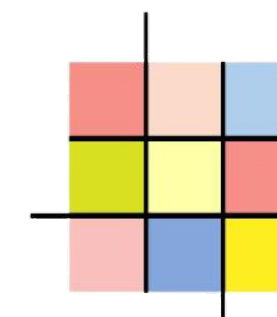


STEM Field: Biology, Computer Science

How does Cannabis affect the brain structure?

Science Snacks for Summer, 17.06.24

Presented by Taeyeon and Zoe



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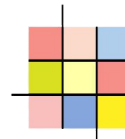
06 CONCLUSION

07 EVALUATION



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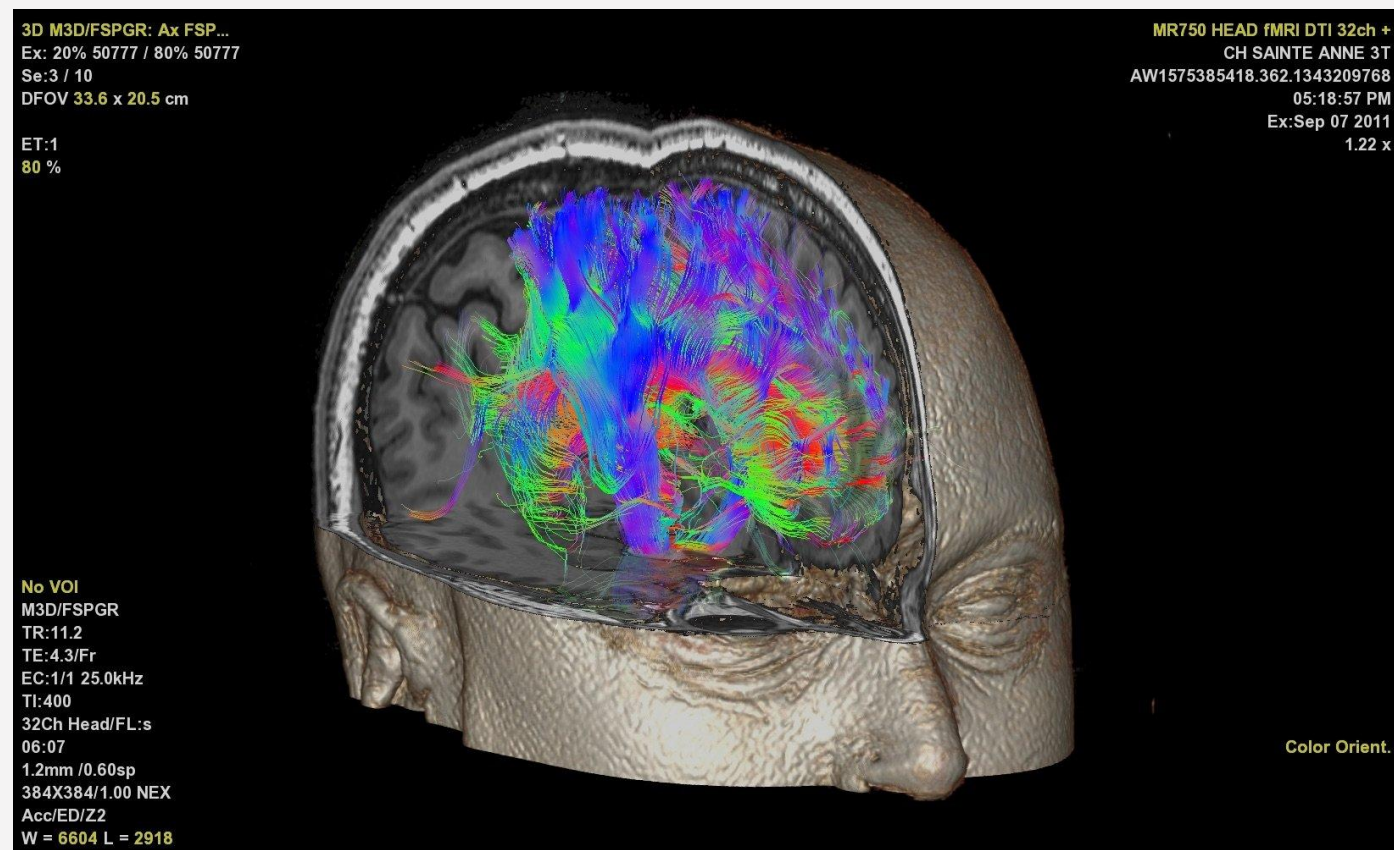
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add statistics - !! how many ppl consume legal in germany!

INTRODUCTION and BACKGROUND

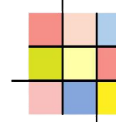
Our research focuses on the dangers of drug addiction and the general effects of cannabis.

Through this research we aim to raise awareness about the dangers of drugs by modelling the usage of cannabis and outlining the specific effects of it.



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Cannabis (Marijuana)

- Methods of consumption
 - Smoking
 - Vaporizing
 - Edibles (food & drinks)
 - Pills
- Benefits:
 - pain relief
 - less anxiety and depression
 - improved sleep
- Some places have fully legalized cannabis use, while others have only legalized medical use, decriminalized possession, or maintained full prohibition



(<https://www.bundesgesundheitsministerium.de/themen/cannabis/faq-cannabisgesetz>)

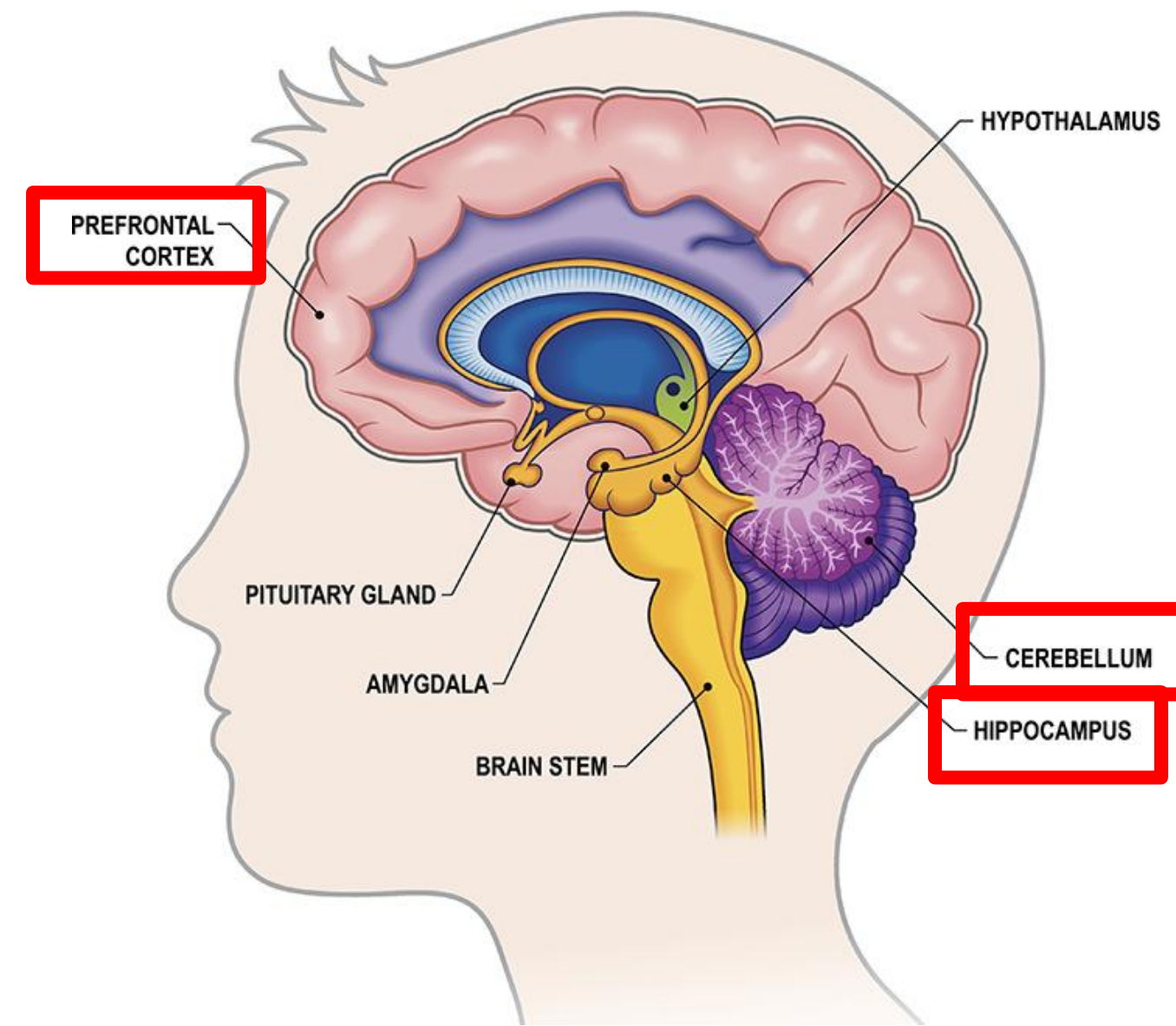
Previous Research

- addiction
- respiratory issues
- cognitive impairment (bad memory, lower concentration levels)
- mental health

- **Effects on brain structure:**

- **hippocampus**
- **striatum**
- **prefrontal cortex**
- **cerebellum**

- change in neurotransmitter systems
- white matter microstructural integrity and gray matter cortical thickness/density
- reduction in overall brain volume
- asymmetrical changes



Methodology

1.

- Research about fMRI
- Analyze normal brain structure
- Research about previous studies

2.

- Find data set
- Modeling
- Test data

3.

- Data Analysis
- Develop Graph

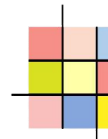
4.

- Conclusion
- Evaluation



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magnetic resonance imaging (MRI)

participants						
	gender	age at baseline	age at onset first CB use	age at onset frequent CB use	audit total baseline	audit total follow-up
	male	25.62	18	n/a	6	3
	ale	18.55	n/a	n/a	2	9
	male	17.66	17	n/a	3	11
	ale	19.75	14	n/a	11	13
	ale	22.2	21	n/a	0	4
	ale	20.95	20	n/a	7	11
	male	20.95	14	n/a	1	3
	ale	25.97	18	n/a	7	6
	ale	20.18	n/a	n/a	2	3
	male	19.88	20	n/a	2	2
	male	18.67	21	n/a	1	4
	male	22.92	21	n/a	3	4
	ale	26.04	n/a	n/a	8	10
	ale	22.83	n/a	n/a	3	5
	female	19.45	14	n/a	8	7
311	HC	male	20.97	n/a	10	11
313	HC	female	21.98	n/a	1	3
314	HC	male	24.63	n/a	1	1
316	HC	male	23.78	n/a	3	8
318	HC	male	21.28	n/a	4	4
319	HC	male	20.37	19	10	7
320	HC	male	19.63	23	4	7
101	CB	male	24.44	13.5	11	10
103	CB	male	24.33	16	22	10
				16	8	13
				14	6	5
			16	18	7	11
			15	17	9	15
				14	3	1
			14	14	2	9
			14	14	11	7
			12	14	6	20
			15	16	7	11
			18	19	7	8
			17	18	5	5
				16	4	9
				14	0	0
			14	15	7	5
			16	16	5	10
			15	16	12	12
			12	13	6	9
			13	18	0	0

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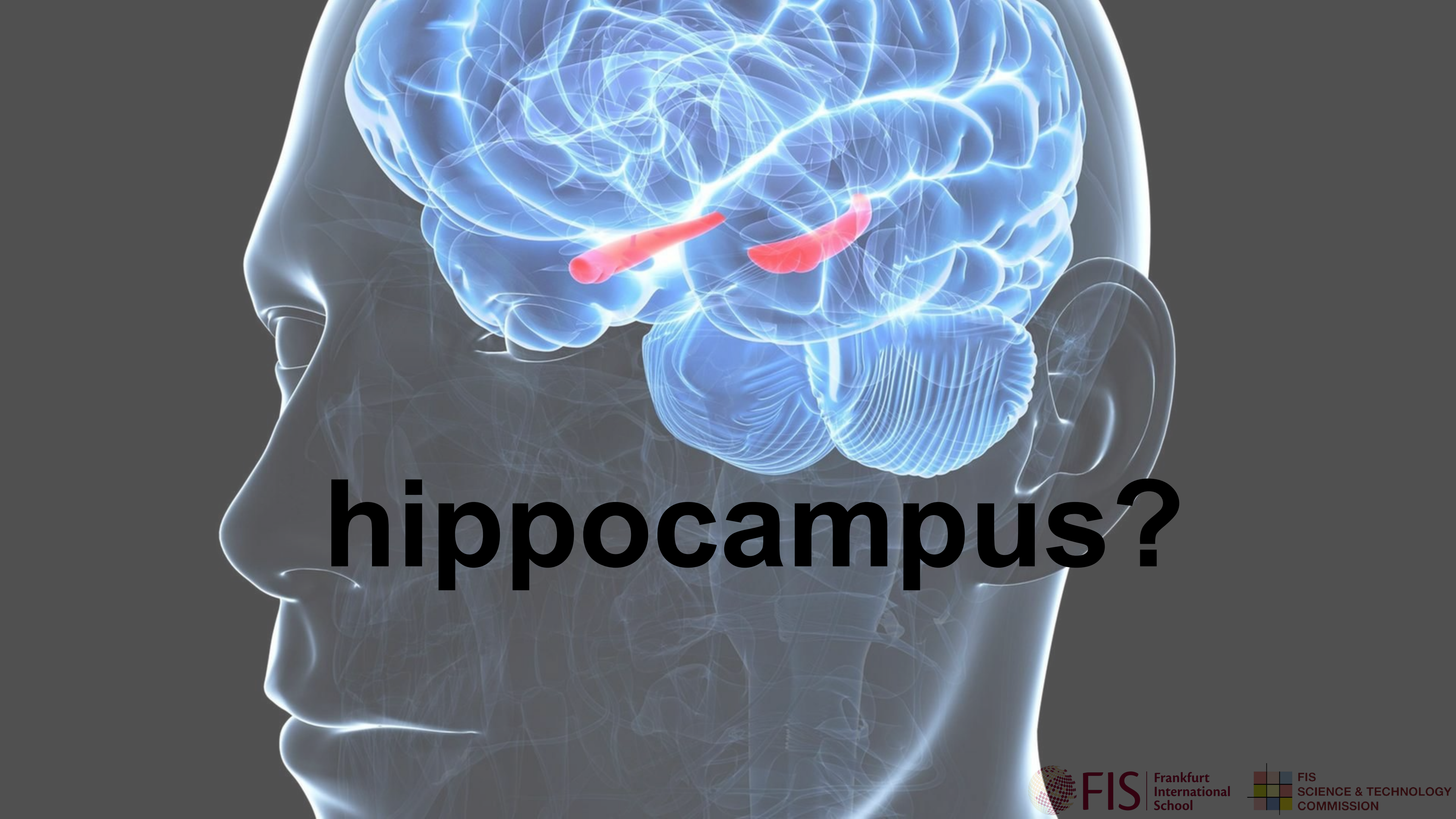


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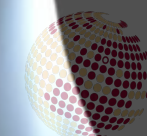
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hippocampus?



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Hippocampus

- embedded deep into temporal lobe
- easily damaged
- hold short-term memories and transfer them to long-term storage in our brains
- emotional processing, including anxiety and avoidance behaviors

308

gender: male

age at

baseline: 23

age at onset

first CB

use: n/a

age at onset

frequent CB

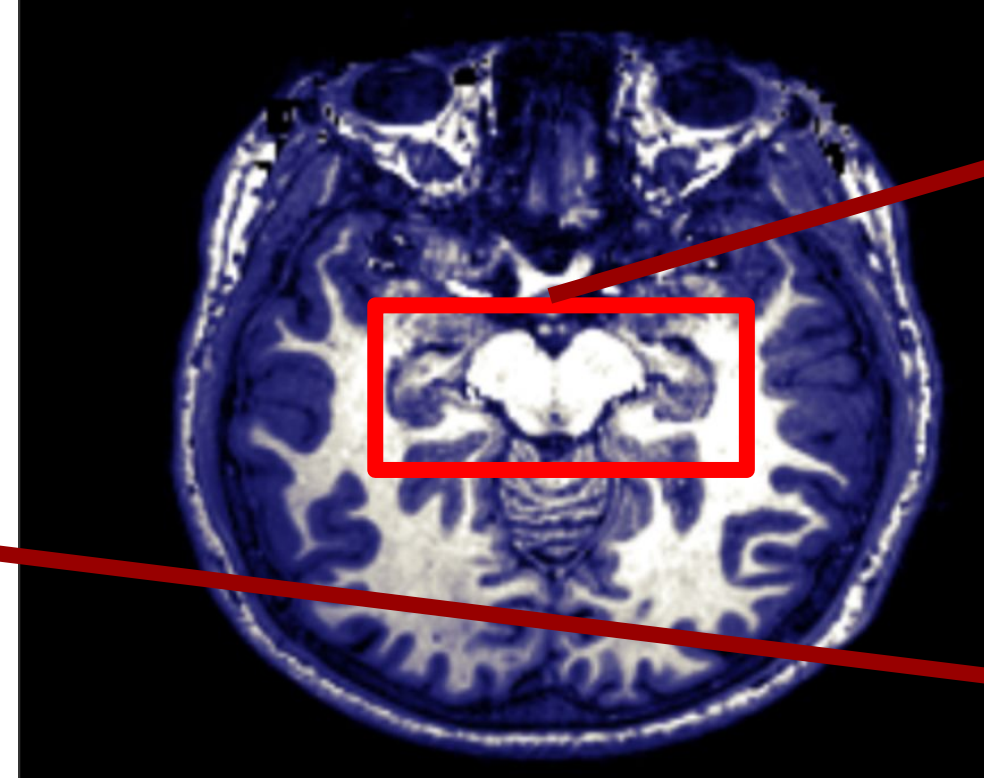
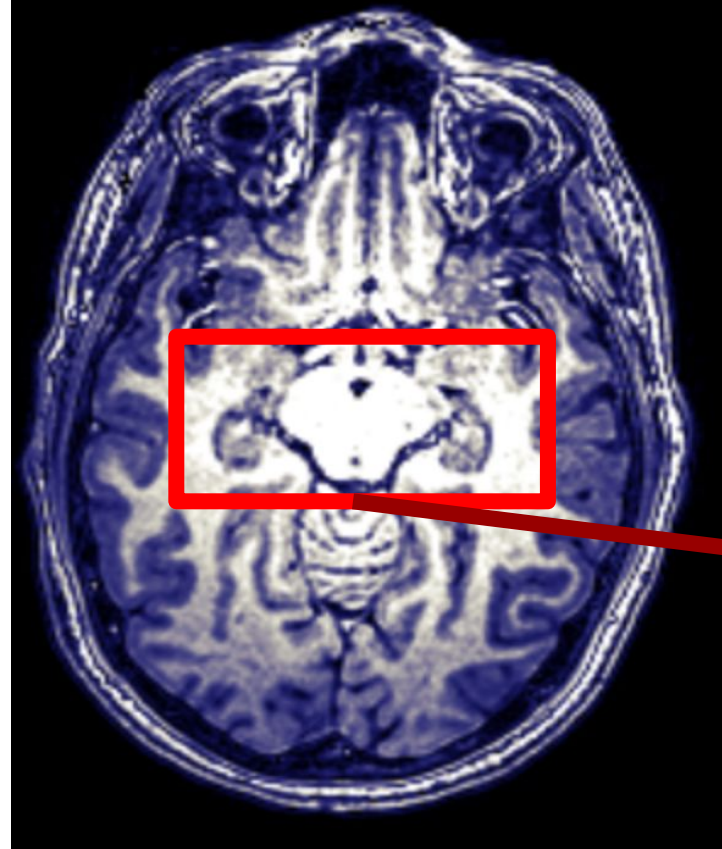
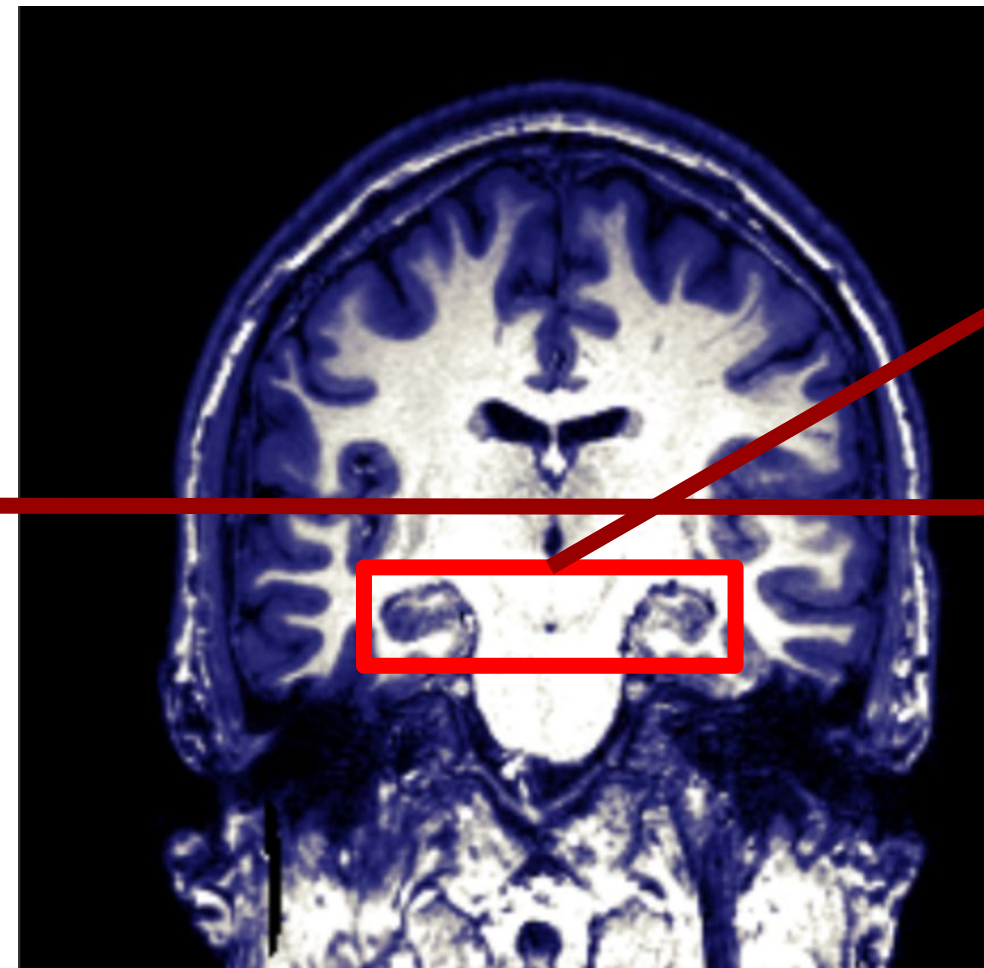
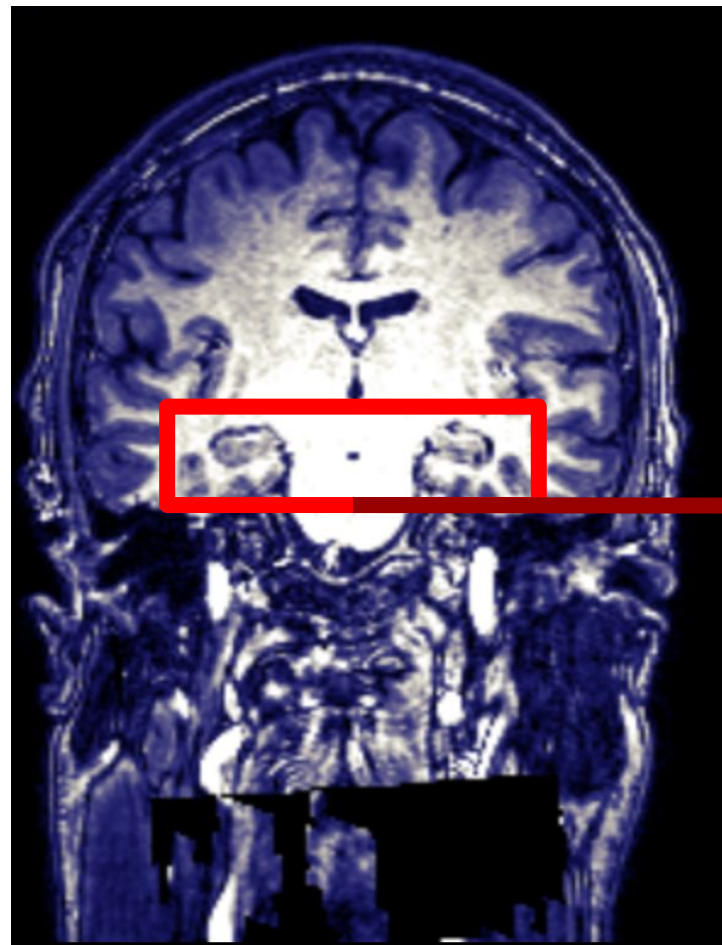
use: n/a

audit total

baseline: 3

audit total

follow-up: 5



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121

gender: male

age at

baseline: 19

age at onset

first CB use: 12

age at onset

frequent CB

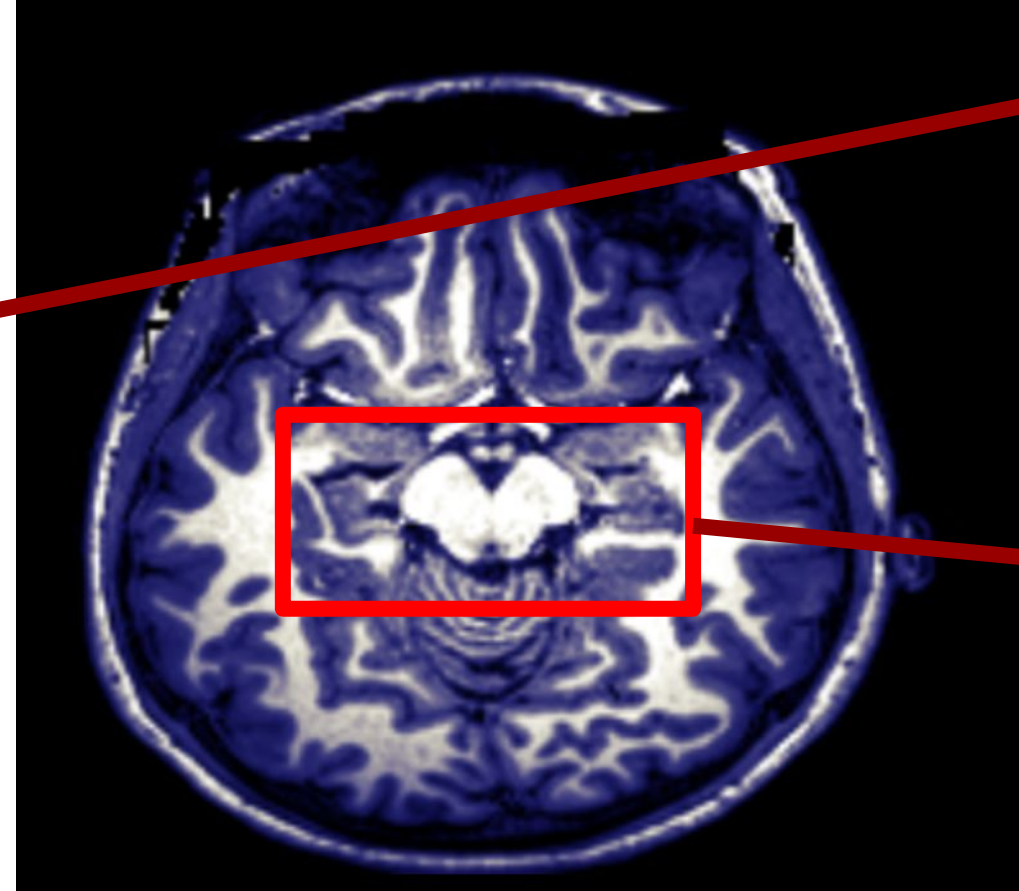
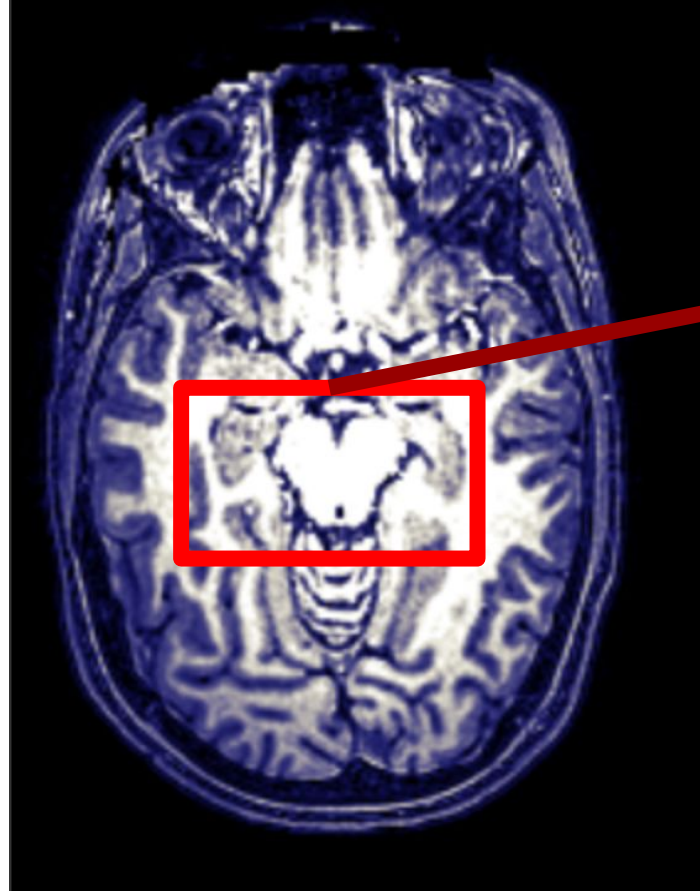
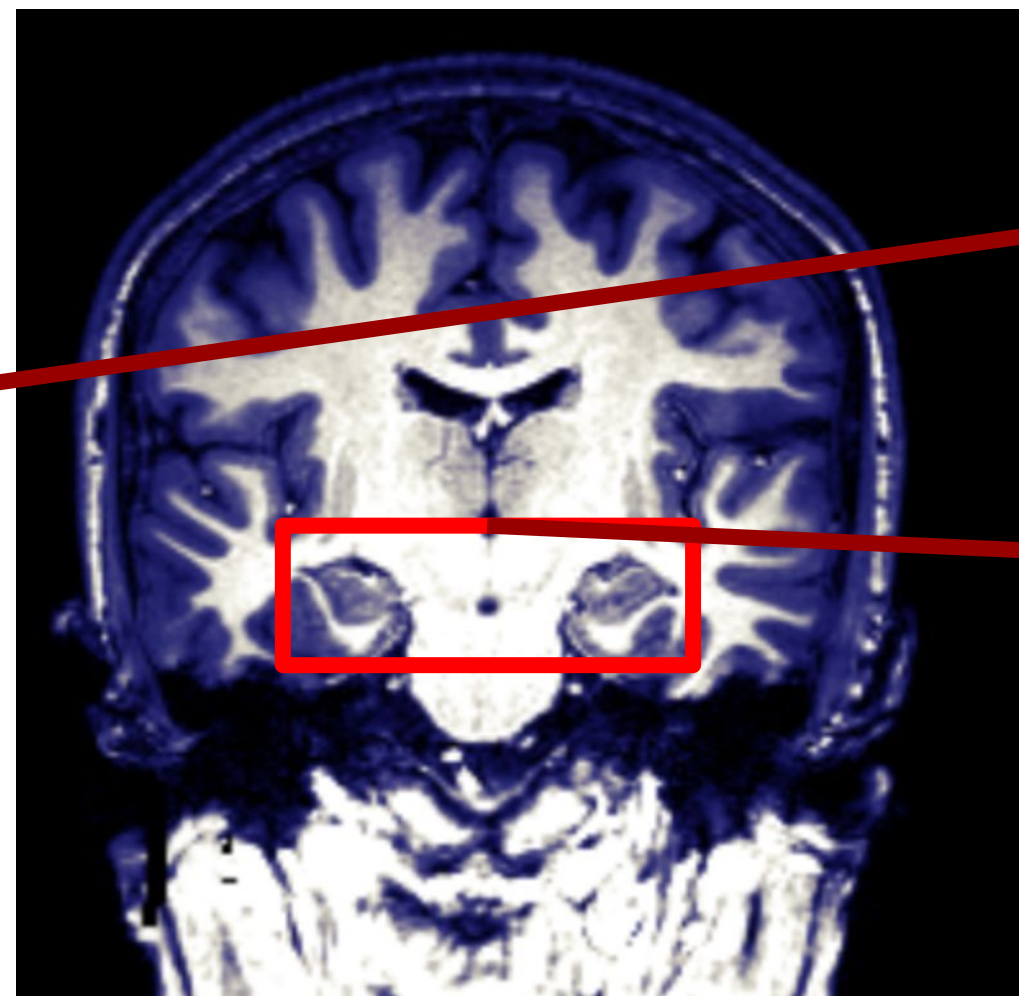
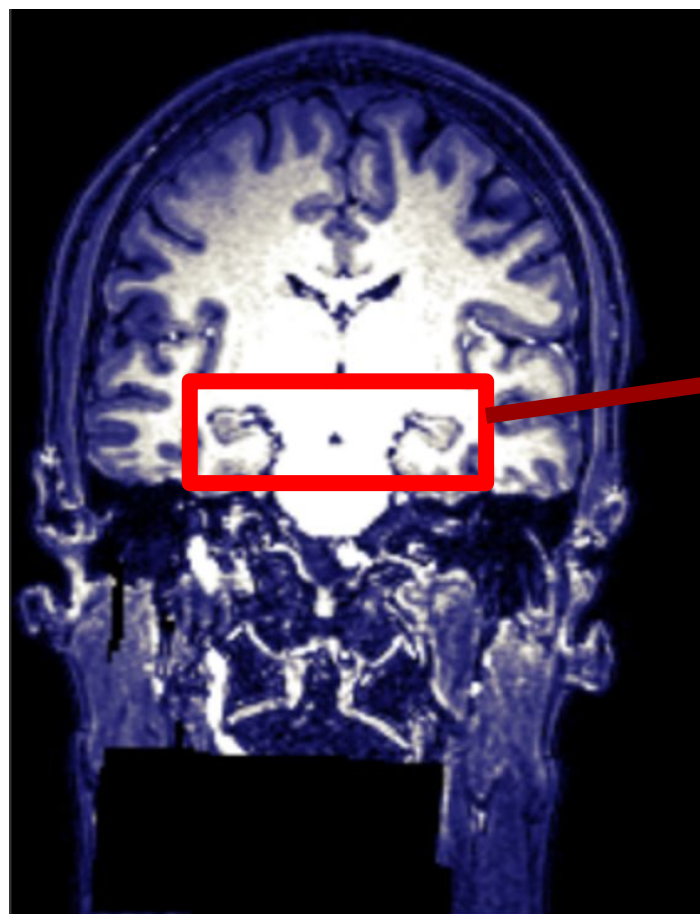
use: 14

audit total

baseline: 6

audit total

follow-up: 20



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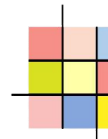


striatum?

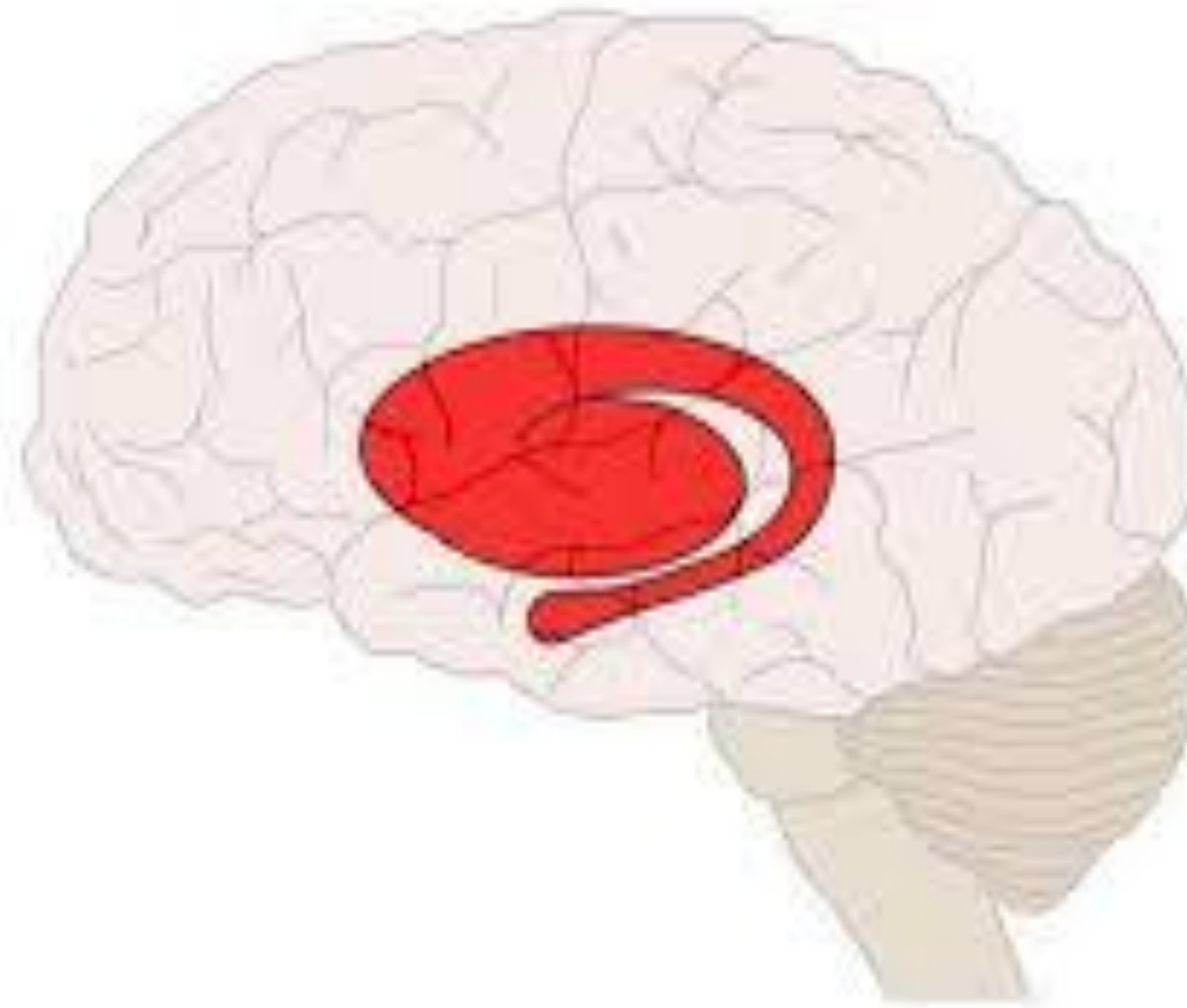


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Striatum

- cluster of interconnected nuclei that form a part of the basal ganglia.
- involved in decision making functions, such as motor control, emotion, habit formation, and reward.

320

gender: male

age at baseline: 20

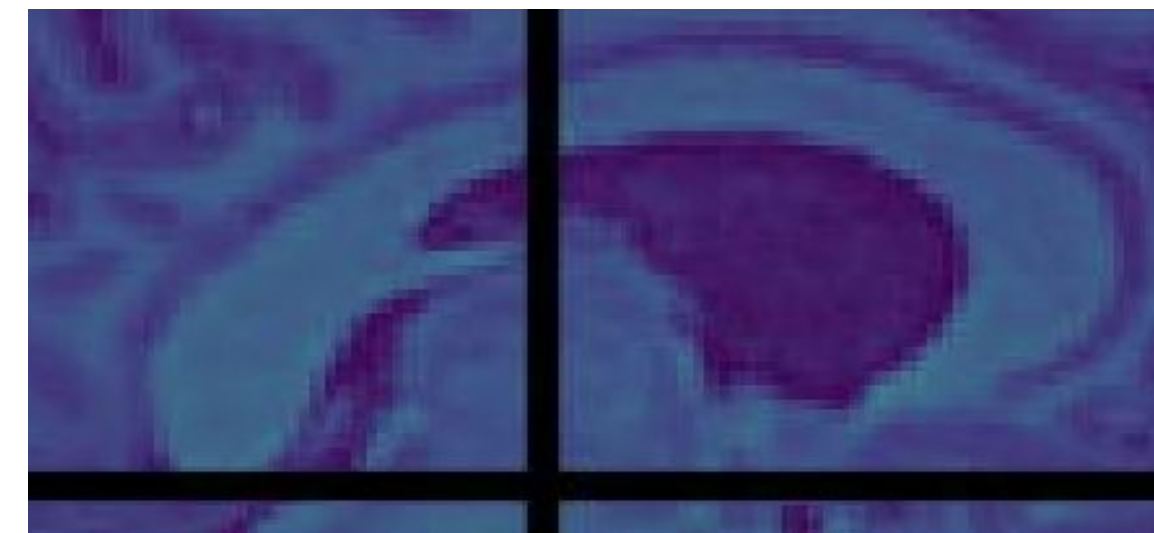
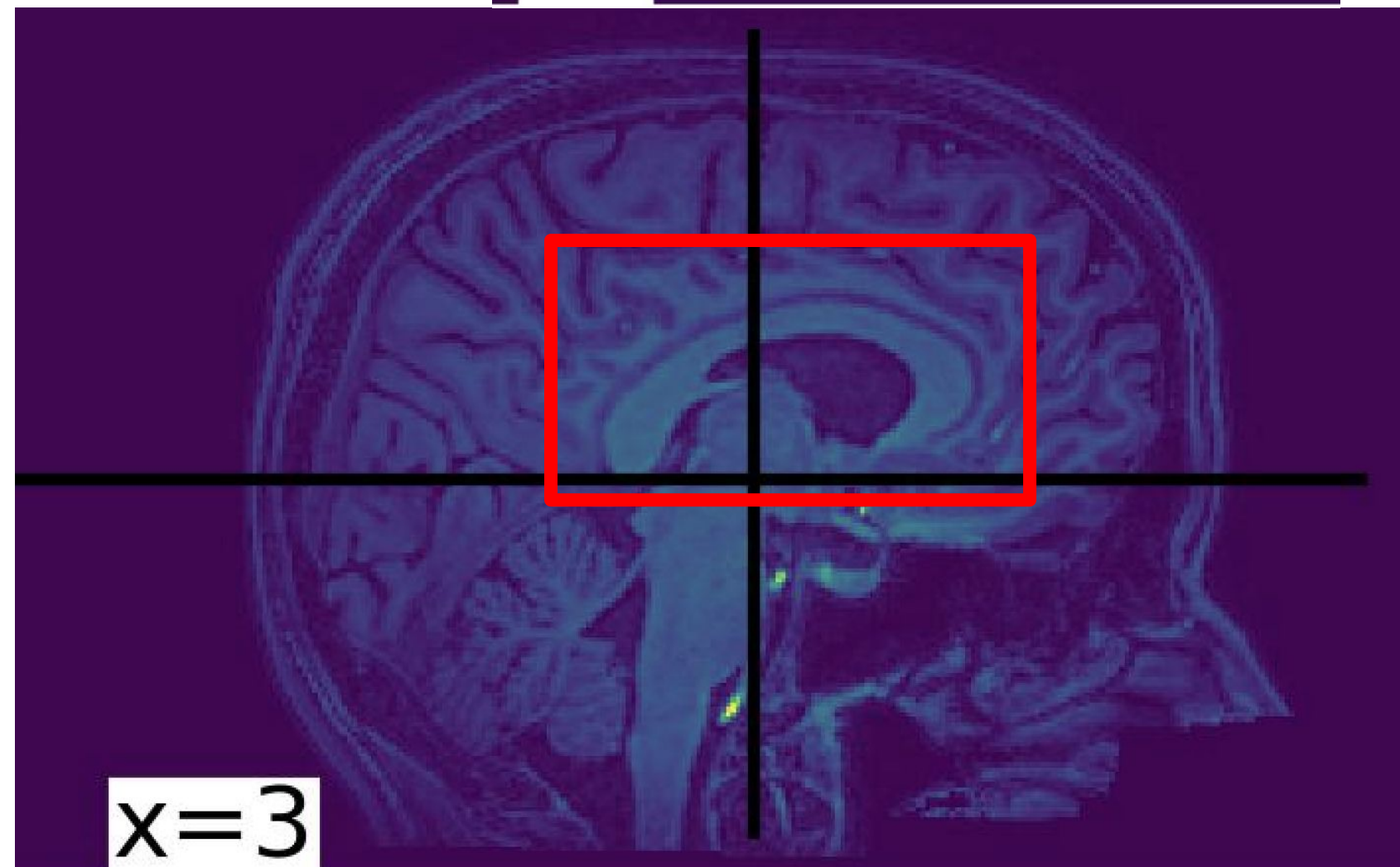
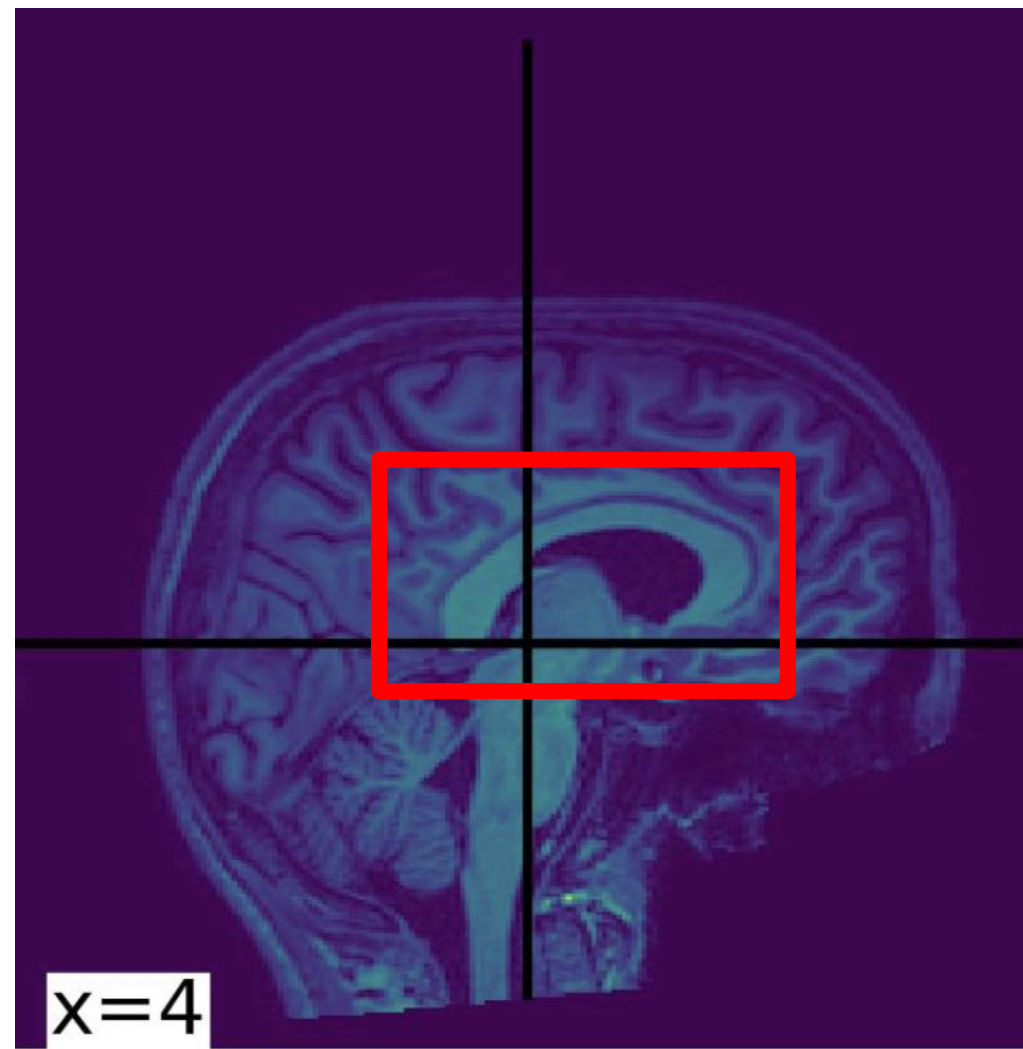
age at onset first CB use: 23

age at onset frequent CB use:

n/a

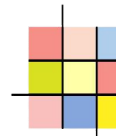
audit total baseline: 4

audit total follow-up: 7



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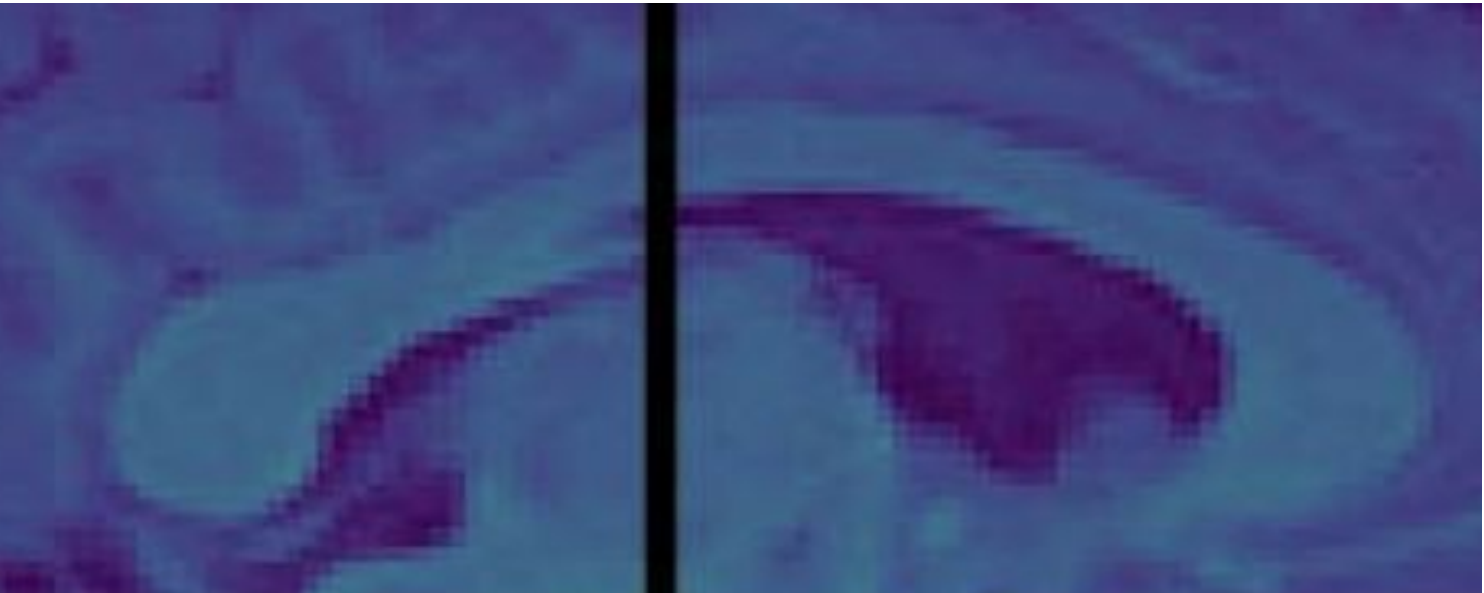
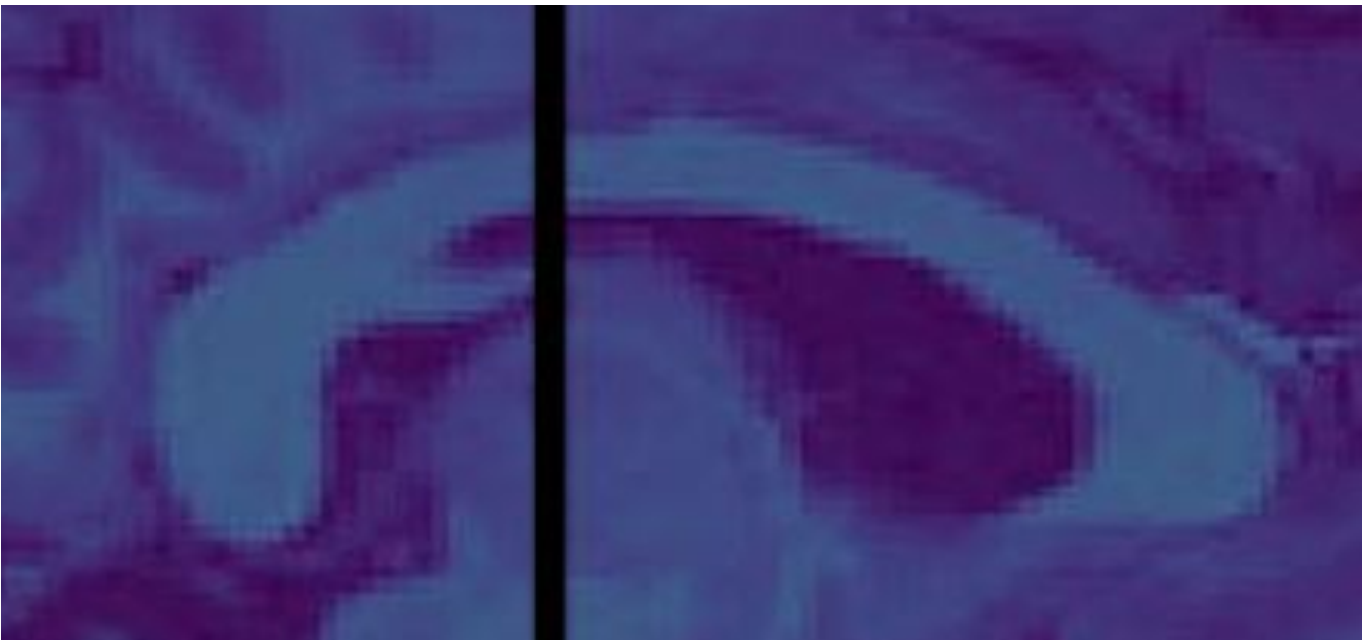
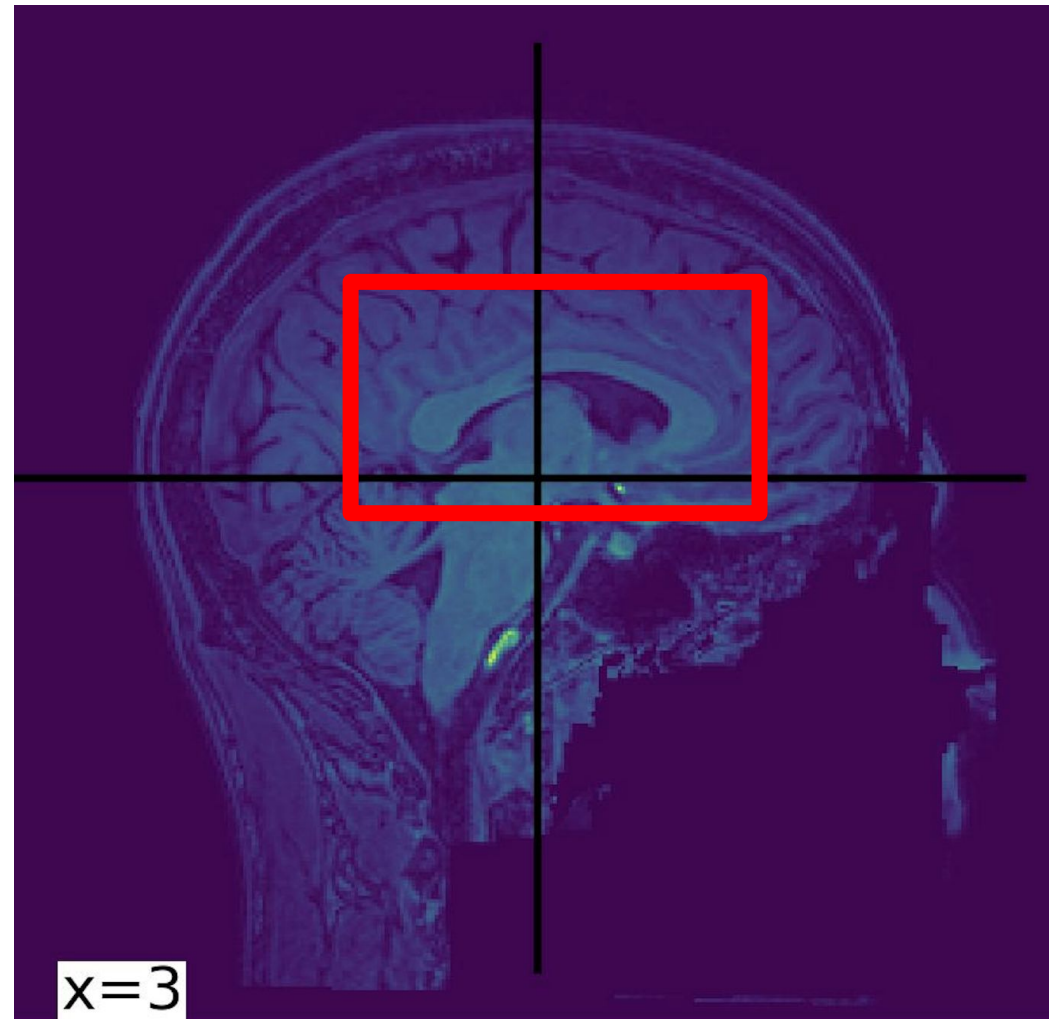
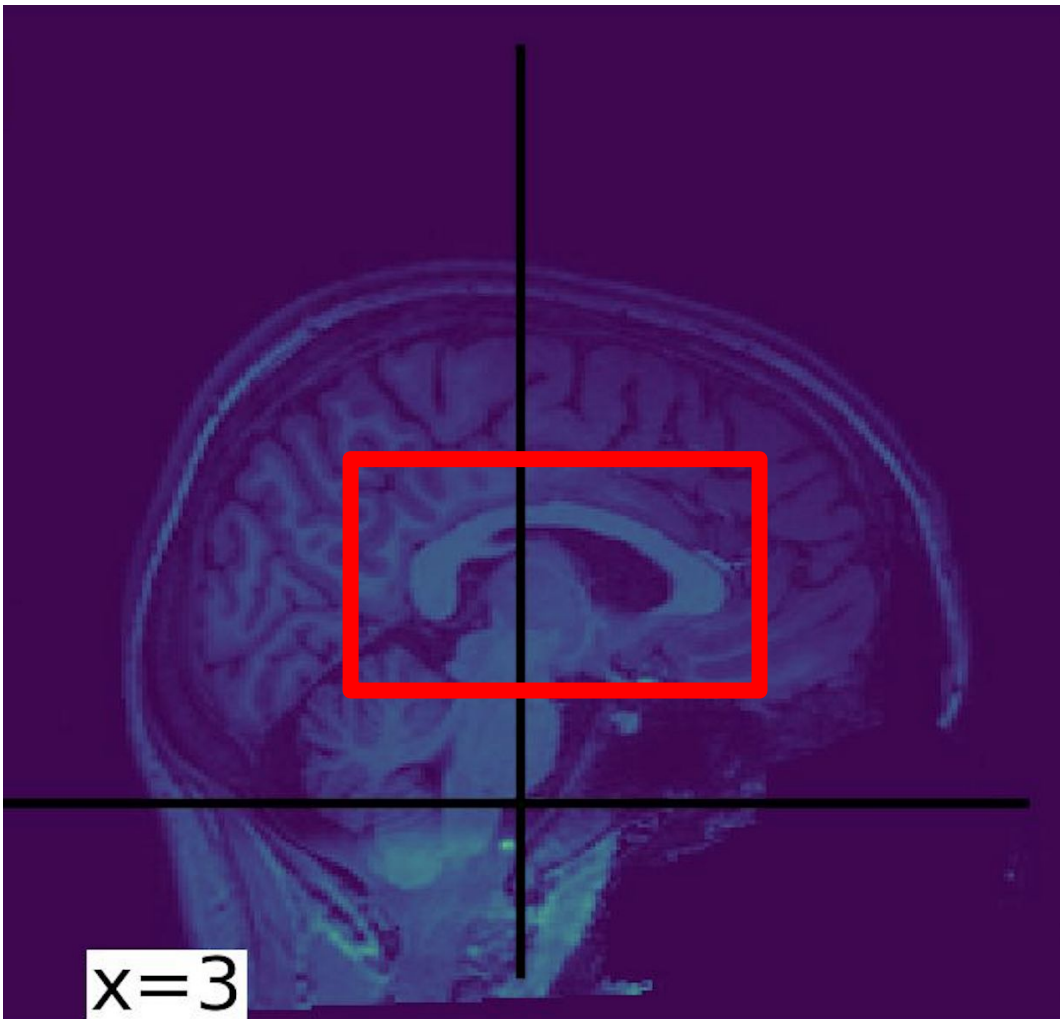
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117

gender: male
age at baseline: 21
age at onset first CB use: 14
age at onset frequent CB use:
14
audit total baseline: 2
audit total follow-up: 9

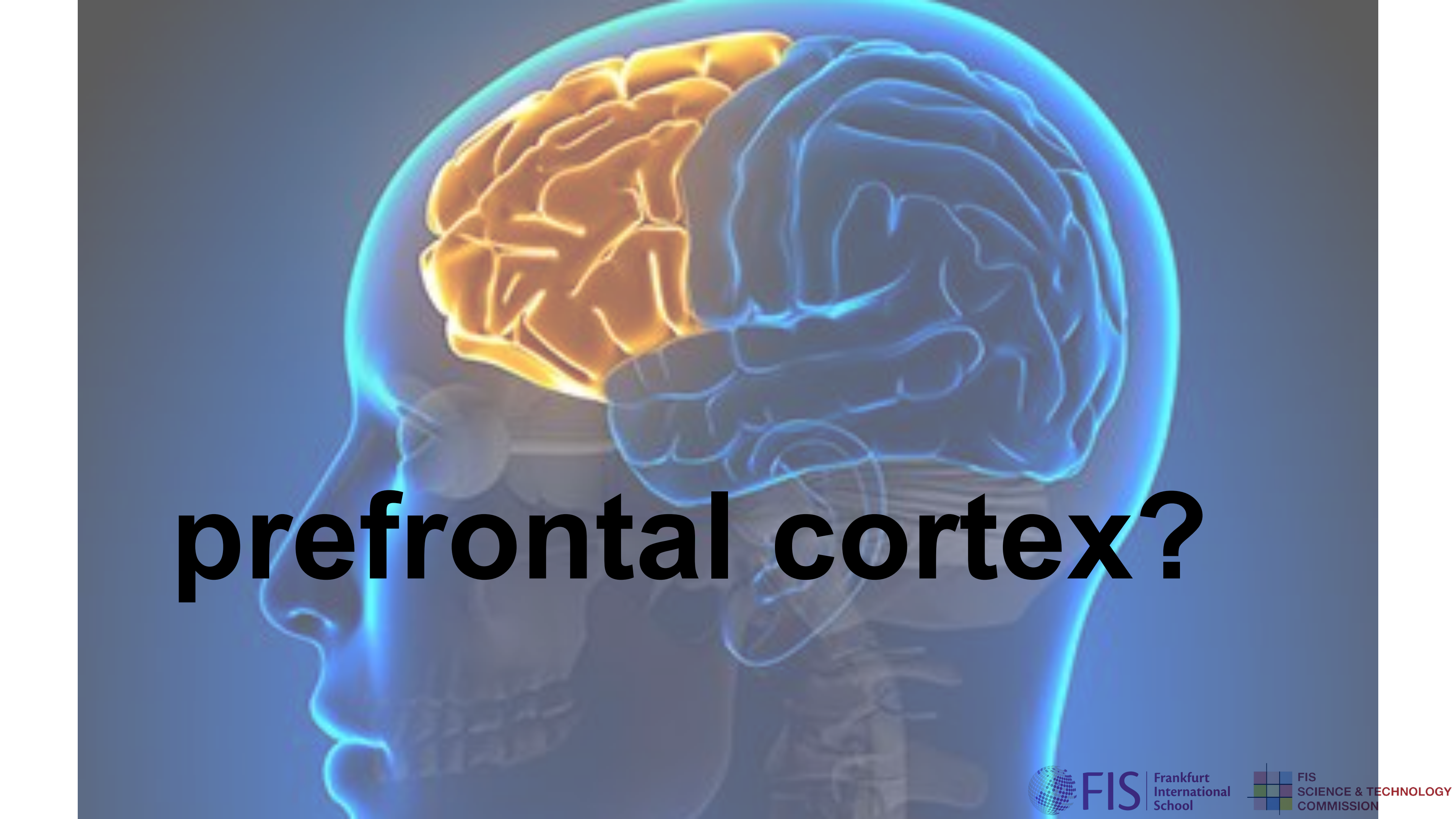


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prefrontal cortex?

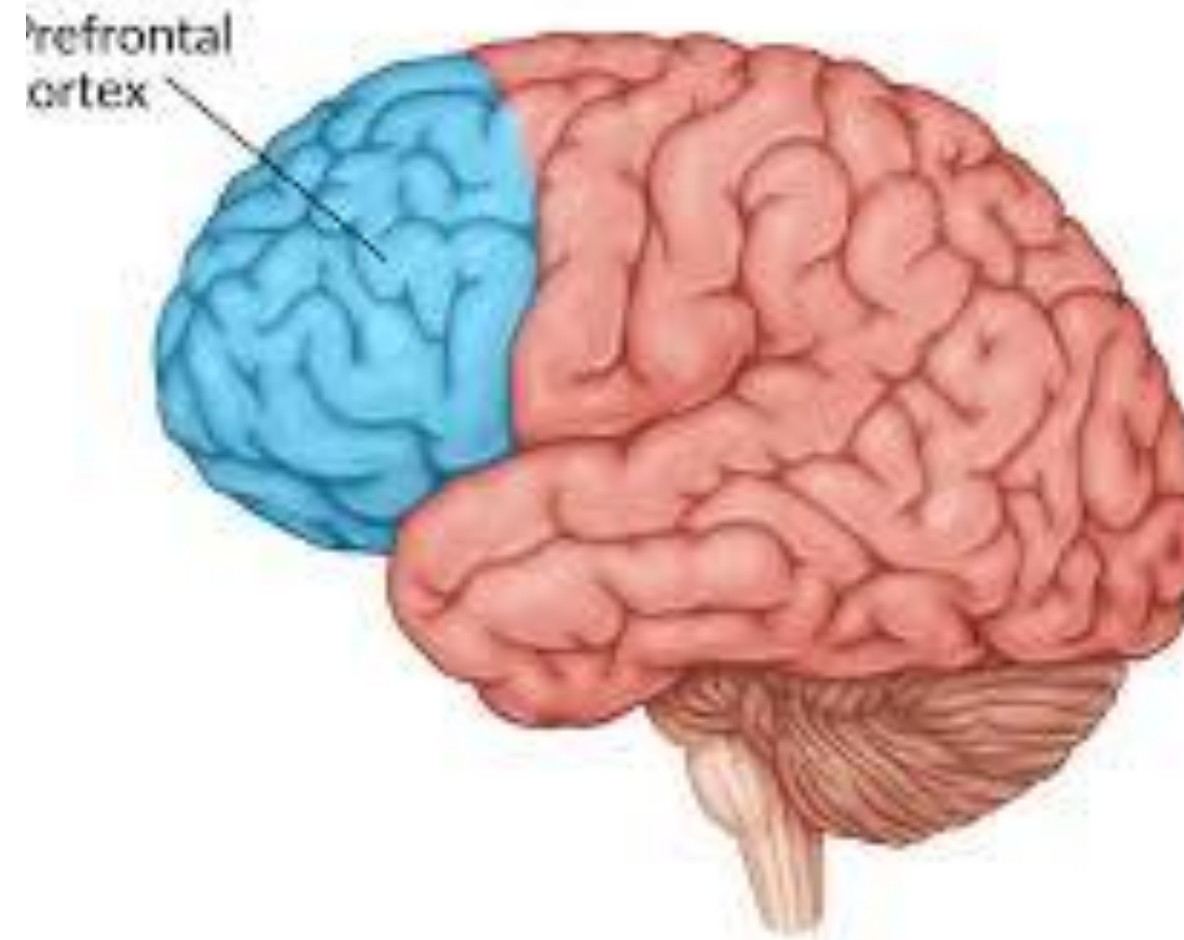


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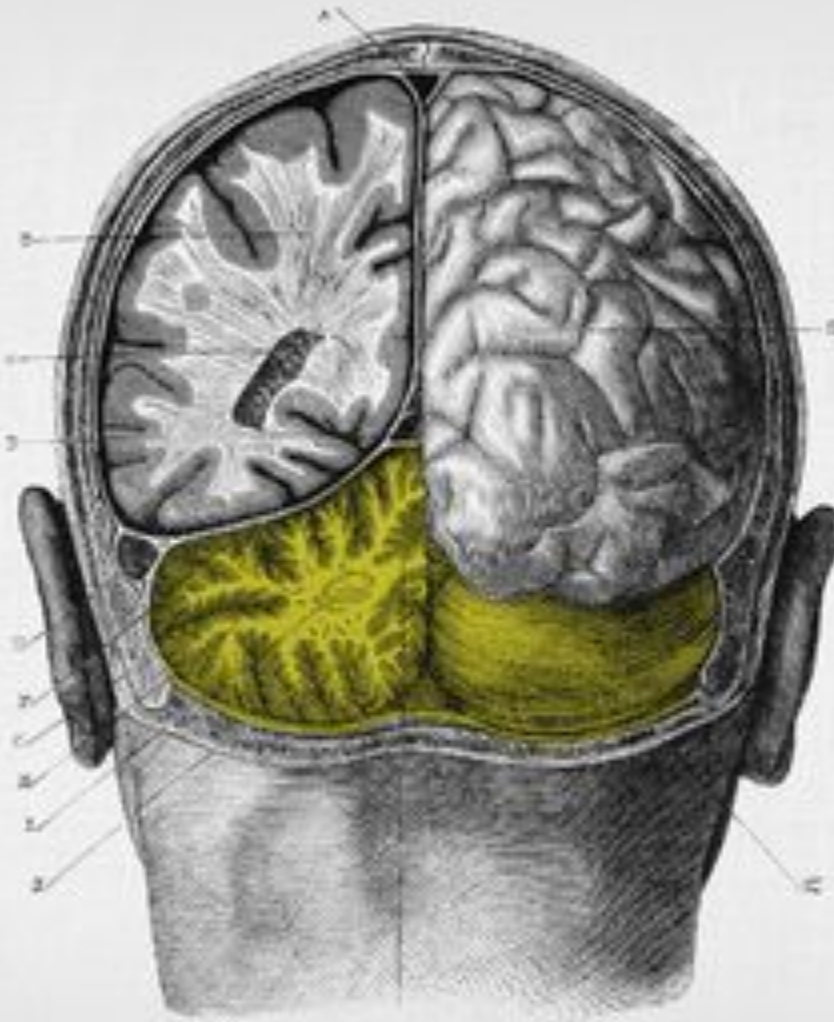
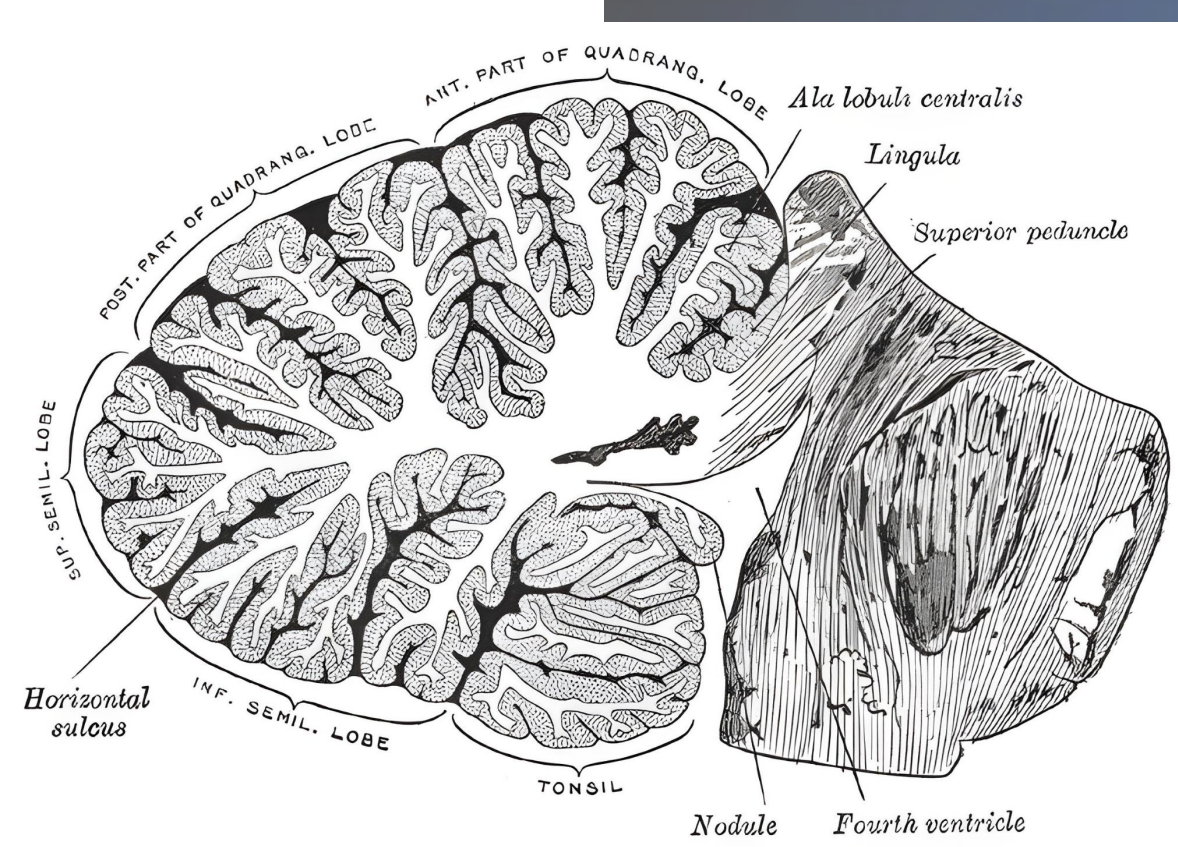
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prefrontal cortex

- regulates our thoughts, actions and emotions through extensive connections with other brain regions
- main connector
- organization via central executive processes
- damaged prefrontal cortex often lead to decreased empathy, short-term memory loss and amnesia

cerebellum?

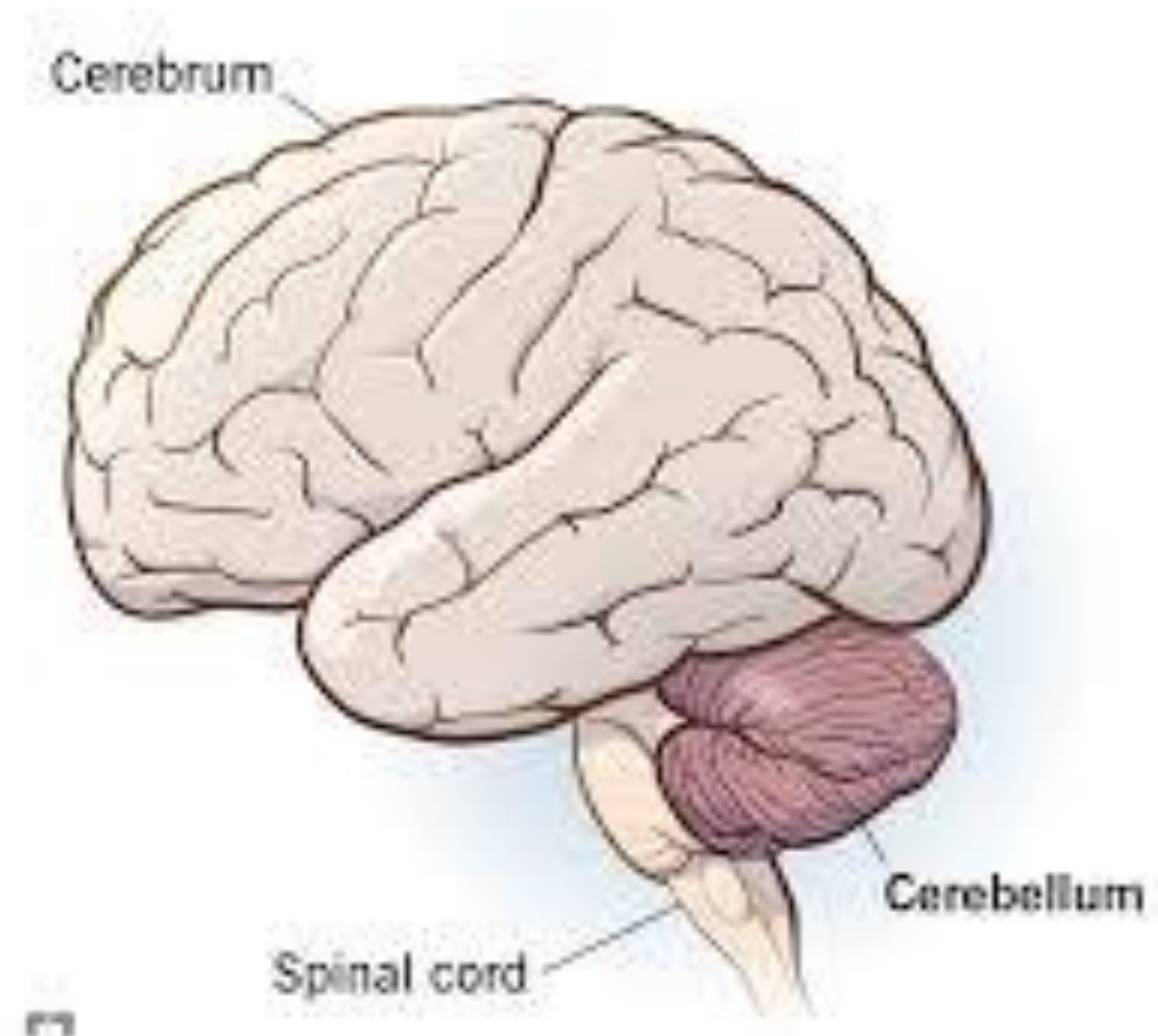


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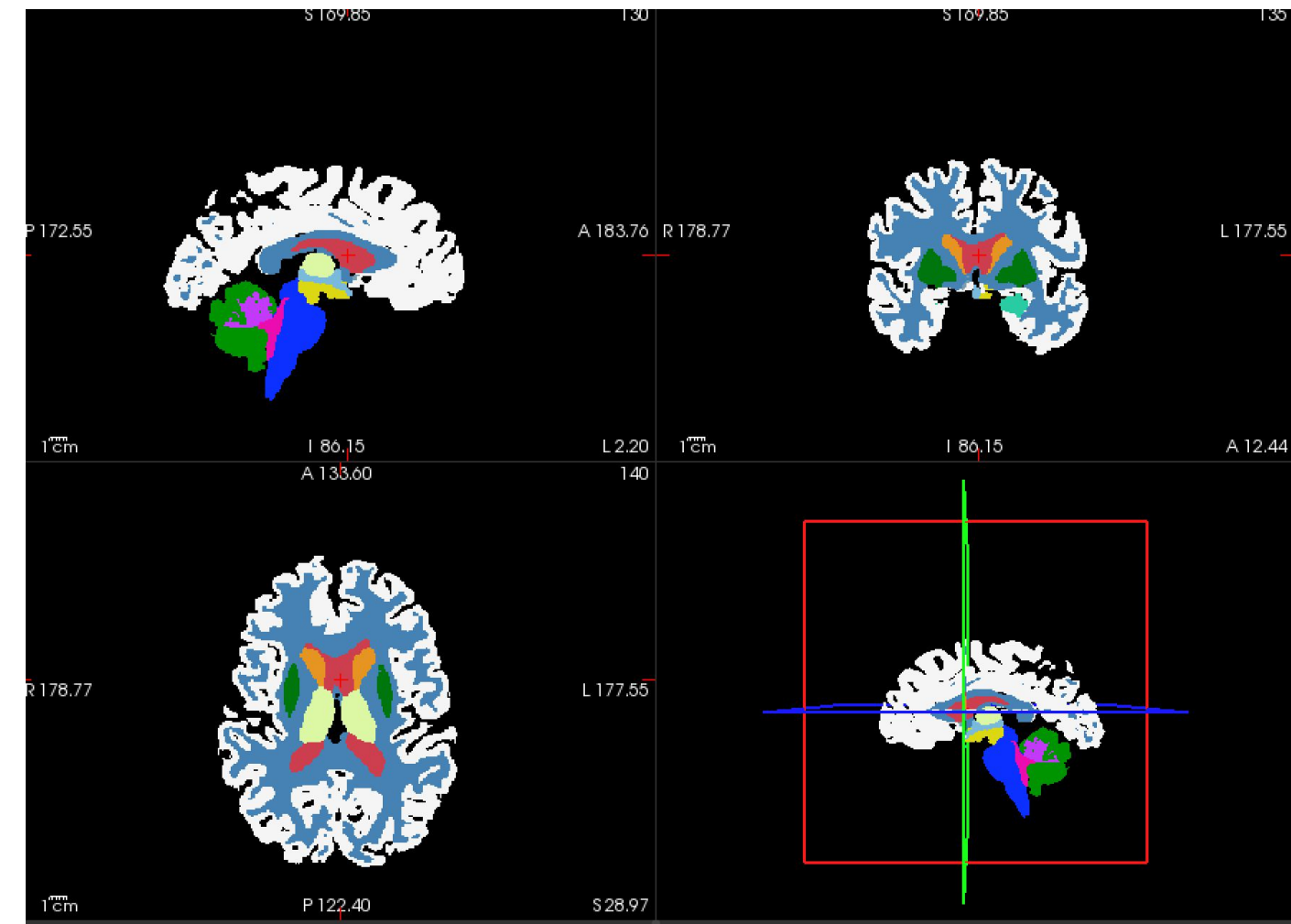


cerebellum

- has special sensors that detect shifts in balance and movement
- sends signals for the body to adjust and move
- Damaged cerebellum causes:
 - loss of coordination of motor movement
 - inability to judge distance and when to stop
 - inability to perform rapid alternating movements

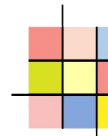
overall brain

volume/intensity

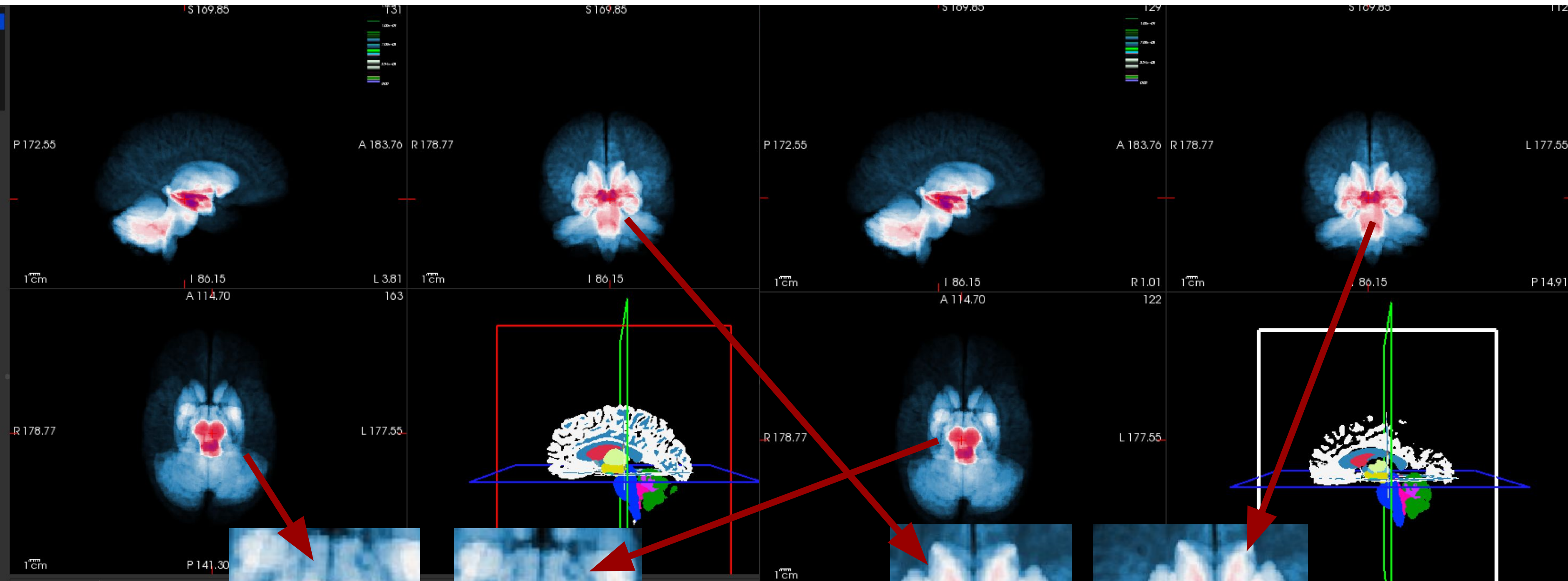


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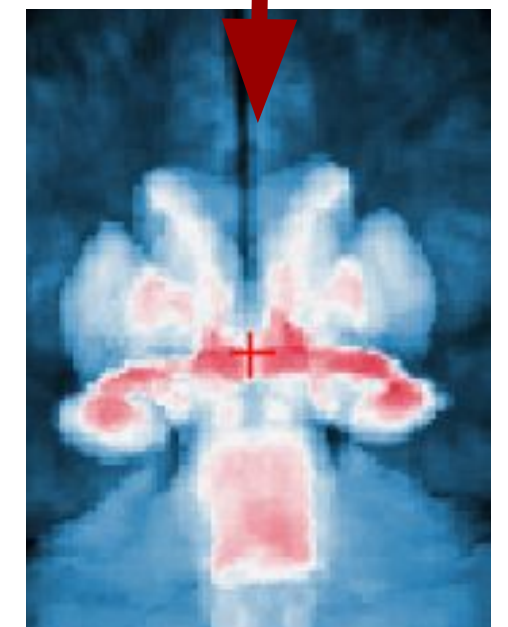
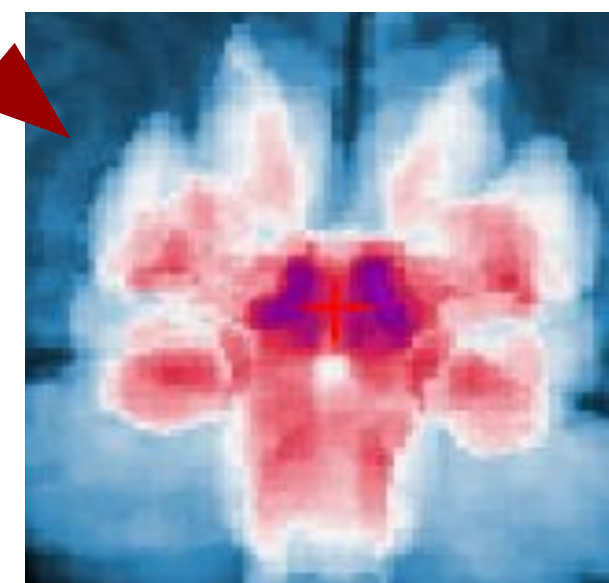
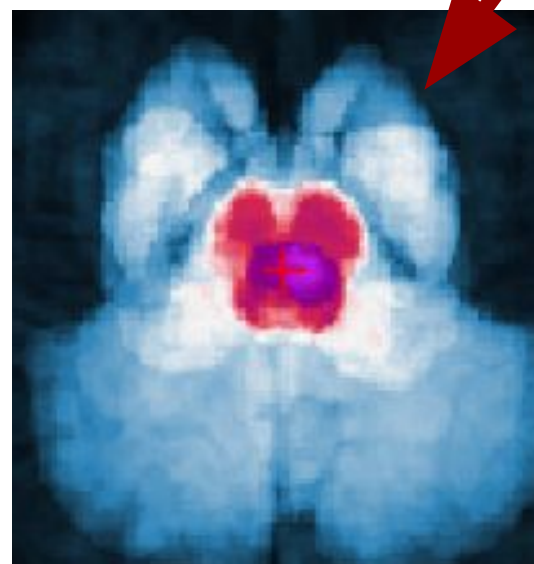
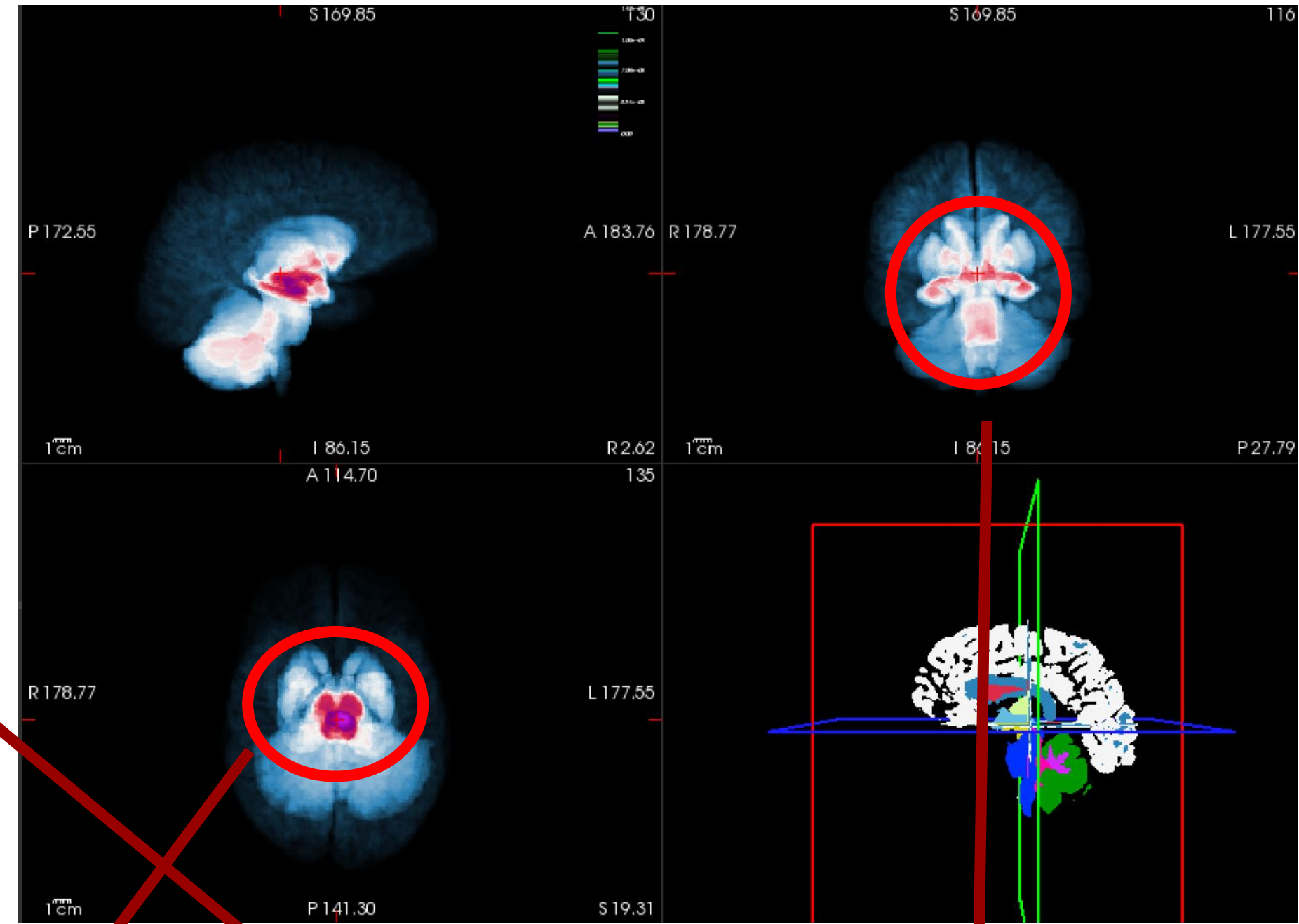
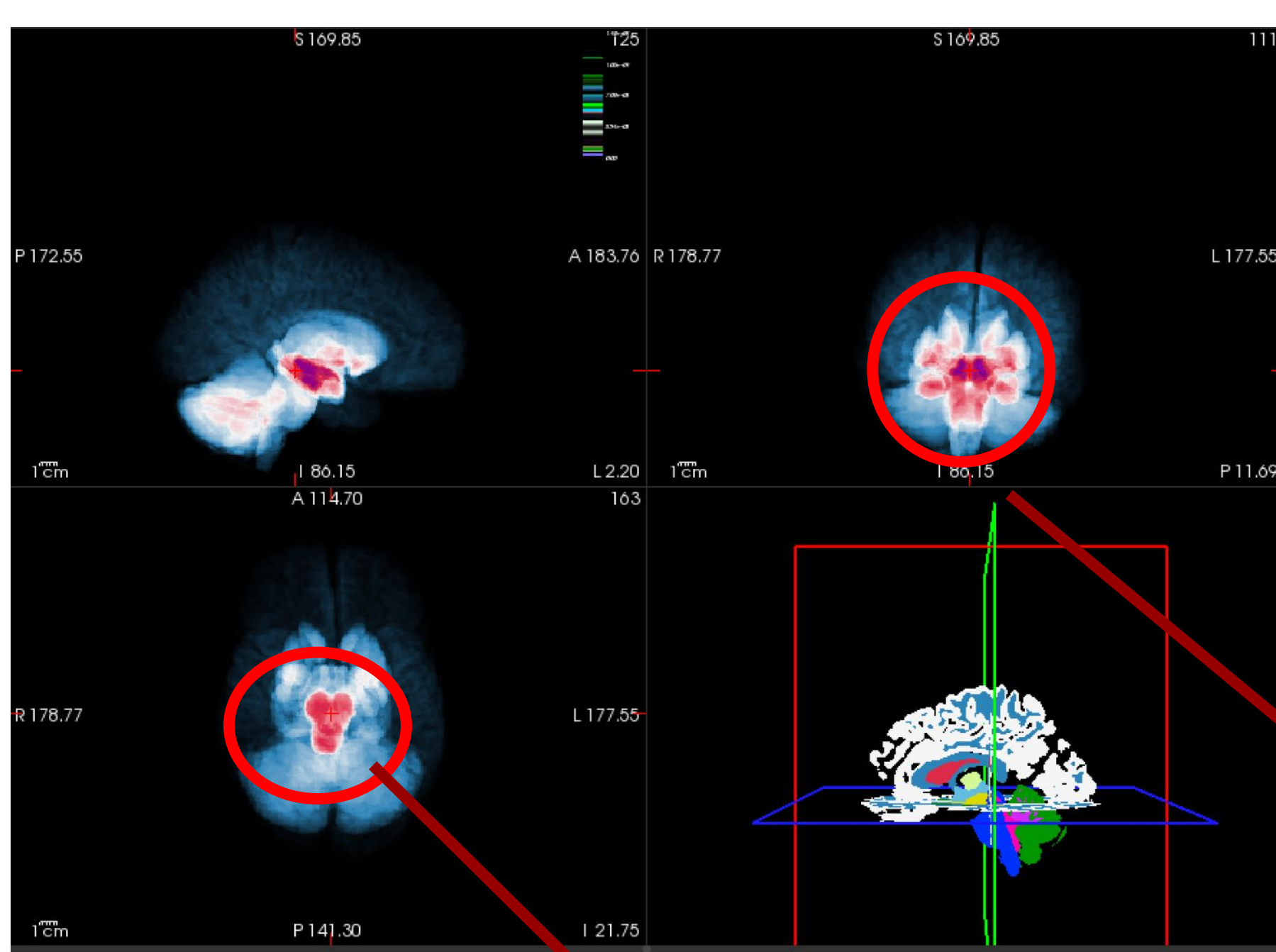


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gender: male
age at baseline: 24
age at onset first
use: n/a
age at onset frequent CB
use: n/a
audit total baseline: 3
audit total follow-up: 8

318



104

gender:mfeale
 age at baseline:18
 age at onset first use:15
 age at onset frequent CB
 use: 16
 audit total baseline: 8
 audit total follow-up: 13



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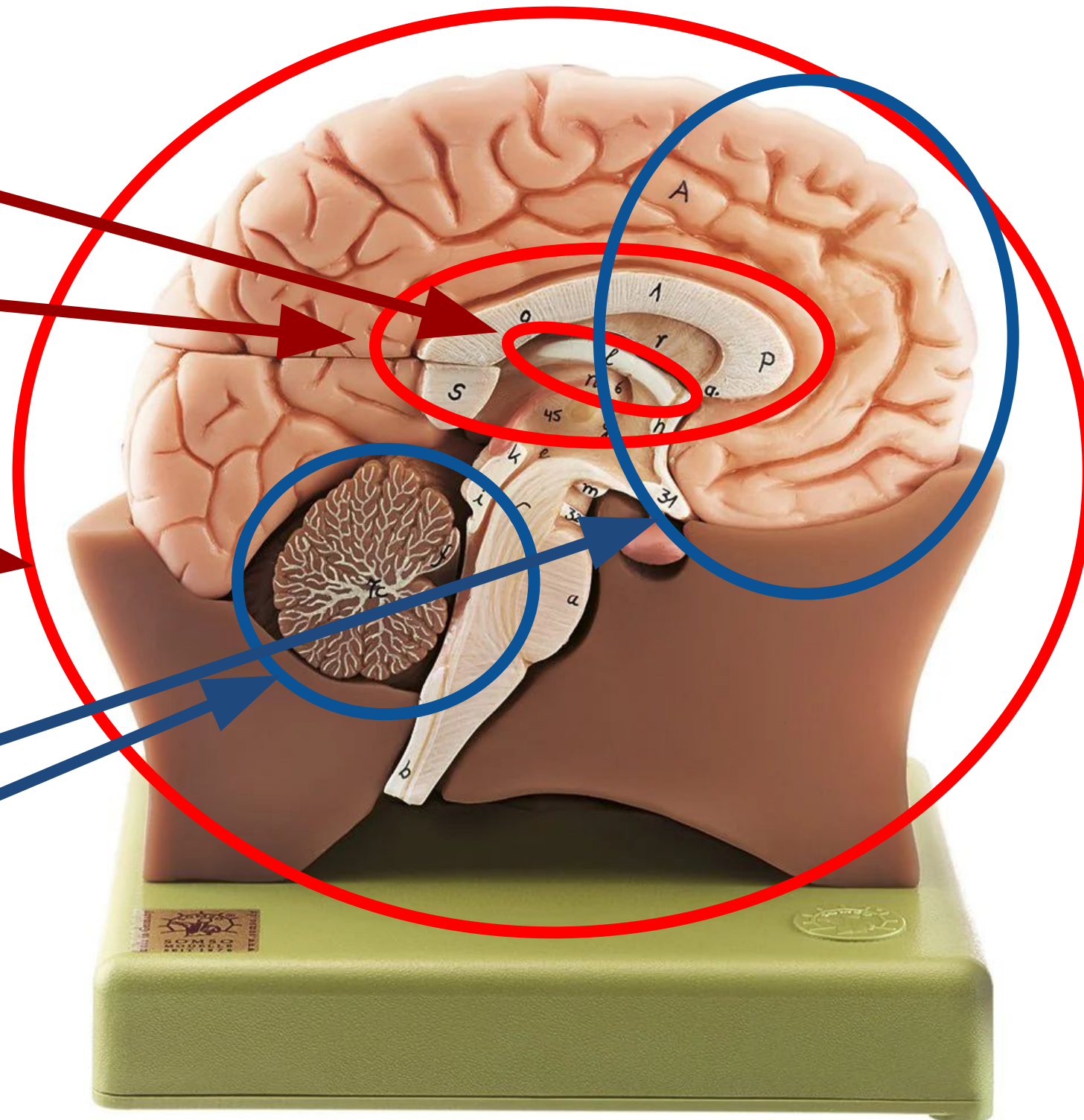
Conclusion

Structural changes observed in

- Hippocampus
- striatum
- overall brain intensity
(/volume)

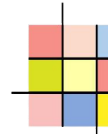
not observed in

- prefrontal cortex
- cerebellum



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Evaluation

STRENGTHS

1. Wide range of resources and previous research
2. Direct comparison between before and after drug consumption
3. Data analysis method: real life humans, not mouse party + large data set of 84 data pieces

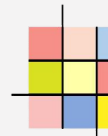
Weaknesses

1. Heavily research-based and hypothetical
2. Other control variables to consider
3. Data is from one source/experiment



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Citations

- <https://www.health.harvard.edu/blog/cognitive-effects-of-long-term-cannabis-use-in-midlife-202206142760#:~:text=Some%20people%20who%20consume%20cannabis,risk%20of%20developing%20cognitive%20symptoms>
- [https://link.springer.com/article/10.1007/s40429-024-00557-z#:~:text=Structural%20MRI,GM\)%20volume%20%5B12%5D](https://link.springer.com/article/10.1007/s40429-024-00557-z#:~:text=Structural%20MRI,GM)%20volume%20%5B12%5D)
- <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7027431/>
- <https://www.medicalnewstoday.com/articles/313265#function>
- <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3548359/#:~:text=Abstract,of%20neurological%20and%20psychiatric%20disorders>.
- [https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2907136/#:~:text=The%20prefrontal%20cortex%20\(PFC\)%20intelligently,brain%20regions%20\(BOX%201\)](https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2907136/#:~:text=The%20prefrontal%20cortex%20(PFC)%20intelligently,brain%20regions%20(BOX%201).).
- <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6656632/#:~:text=The%20striatum%20is%20a%20cluster,%20C%20habit%20formation%20and%20reward>.
- <https://www.webmd.com/brain/hippocampus-what-to-know>
- <https://my.clevelandclinic.org/health/body/23418-cerebellum>
- <https://www.bundesgesundheitsministerium.de/themen/cannabis/faq-cannabisgesetz>
- <https://www.ncbi.nlm.nih.gov/books/NBK425767/>
- <https://www.technologynetworks.com/neuroscience/articles/gray-matter-vs-white-matter-322973>
- <https://www.spinalcord.com/blog/gray-matter-vs-white-matter-in-the-brain>

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of cannabis

Thank you for your attention

Any Questions?

Presented by Taeyeon Kim and Zoe Zhang